Question:

The y-intercepts, P and Q, of two perpendicular lines intersecting at the point $A(6,8)$ have a sum of zero. What is the area of $\triangle APQ$?
$\textbf{(A)}\ 45\qquad\textbf{(B)}\ 48\qquad\textbf{(C)}\ 54\qquad\textbf{(D)}\ 60\qquad\textbf{(E)}\ 72$

Reasoning + Answer:

<think> To solve the problem, we first need to determine the slopes of the lines. Let the equations of the lines be \(y = m(x-6) + 8\) and \(y = -\frac{1}{m}(x-6) + 8\), where the lines are perpendicular and intersect at \(A(6,8)\). The y-intercepts of these lines are \(P(0,8+6m)\) and \(Q(0,8-6/m)\), and the sum of the y-intercepts is zero, so we have:
\[8 + 6m + 8 - \frac{6}{m} = 0\]
\[16 + 6m - \frac{6}{m} = 0\]
Multiplying through by \(m\) to clear the fraction:
\[16m + 6m^2 - 6 = 0\]
\[6m^2 + 16m - 6 = 0\]
Dividing the entire equation by 2:
\[3m^2 + 8m - 3 = 0\]
Solving this quadratic equation using the quadratic formula \(m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}\), where \(a = 3\), \(b = 8\), and \(c = -3\):
\[m = \frac{-8 \pm \sqrt{8^2 - 4 \cdot 3 \cdot (-3)}}{2 \cdot 3} = \frac{-8 \pm \sqrt{64 + 36}}{6} = \frac{-8 \pm \sqrt{100}}{6} = \frac{-8 \pm 10}{6}\]
This gives us two solutions for \(m\):
\[m = \frac{2}{6} = \frac{1}{3} \quad \text{or} \quad m = \frac{-18}{6} = -3\]
We will use \(m = \frac{1}{3}\) and \(m = -3\). The equations of the lines are:
\[y = \frac{1}{3}(x-6) + 8 = \frac{1}{3}x - 2 + 8 = \frac{1}{3}x + 6\]
\[y = -3(x-6) + 8 = -3x + 18 + 8 = -3x + 26\]
The y-intercepts are \(P(0,6)\) and \(Q(0,26)\). The coordinates of \(P\) and \(Q\) are \(P(0,6)\) and \(Q(0,26)\). The base of the triangle \(APQ\) is the vertical distance between \(P\) and \(Q\), which is \(26 - 6 = 20\). The height of the triangle is the horizontal distance from \(A\) to the y-axis, which is 6. The area of \(\triangle APQ\) is:
\[\text{Area} = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 20 \times 6 = 60\]
</think>
<answer>60</answer>